# Ishaan Shrivastava

Noida, Uttar Pradesh
201301

Links: **Github** **LinkedIn** **Codeforces**
**Portfolio-Website**
**ishaanshri95@gmail.com**

## *Education*

**Indian Insititue of Technology, Jodhpur** November '20 – May '24
*B.Tech in Artificial Intelligence and Data Science - CGPA 8.1 / 10* *Jodhpur, Rajasthan, India*

## *Work Experience*

**Computer Vision Research Engineer** May '24 – Current
*Metafusion [Supervisor: Manish Soni]* *Noida, India*

- **Dashcam Monitoring System for NHAI (National Highway Operations &Management)** Developed extremely resource-efficient comprehensive AI analytics backend which includes detection of potholes, rutting, and severe cracking, as well as damaged or faded lane markings, among 35 total categories of road defects. **Deployed across 40,000 kilometres of national highways.**
- Boosted person-detection pretraining over SOTA by 2% with a scalable dataset curation, aggregation and selection pipeline. The improved object detecter obtained a **60% speedup over the previous solution**. Implemented custom data pipeline on an ultralytics fork allowing for **15x increased throughput in training runs** on available hardware, allowing thousands of iterative ablations within a single month.
- Accelerated VLM deployment via **data-free structured pruning** method for attention-layer skipping in VLMs, based on **heavy-tailed self-regularization theory (HTSR)**

**Open source engineer** May '23 – Aug '23
*CodeForGovTech '23 [Supervisor: Gautam Rajeev]* *Remote*

- **shrivastava95/docparser** Built a CLIP-guided language routing system for Tesseract OCR, reducing language selection errors from 13% to 2% on multilingual documents. Open-Sourced docparser, a multilingual document parsing system supporting 116 languages and 37 scripts.
- **Samagra-Development/ai-tools (PR #102)** Dictionary-based augmentations for Azure Odia translation, improving translation robustness in transformer-based models.
- Collaborated with supervisors for successful deployment of **shrivastava95/docparser**, positively impacting and **deployed across millions of farmers** to provide language translations in regional dialects.

**Undergraduate researcher** May '23 – Aug '23
*Supervisor: Dr. Shweta Jain* *Ropar, Punjab, India*

- Improving fairness in non-i.i.d federated classification settings using knowledge distillation, EMA, and stochastic weight averaging. Rewrote the Ditto **federated learning** framework, from TensorFlow to PyTorch, ran ablations achieving an improvement of 0.2%.

## *Projects*

**SparseRNN**
*Supervisor: Prof. Richa Singh, Mayank Vatsa*

- **GitHub shrivastava95/sparsernn** Implemented SparseRNN, an RNN with exponentially dilated temporal connectivity with subquadratic complexity, as an alternative to dense transformer decoder.
- Showed pareto improvements over LSTM, RNN, and dense transformer baselines on three tasks - Sequential MNIST classification, IMDB sentiment classification, Penn Treebank Part-of-Speech tagging

**Hierarchical Reasoning Model (HRM)**
*Supervisor: self*

- **GitHub shrivastava95/HRM-minimal** Minimal implementation of Hierarchical Reasoning Models, a cognition-inspired novel recurrent architecture that executes sequential reasoning tasks like Sudoku and ARC-AGI in a single forward pass without explicit supervision of the intermediate process, beating LLMs more than 1000x larger such as Claude Sonnet 3.7, OpenAI o3-mini, etc.

**DevRev Expert Answers in a Flash: Improving Domain-Specific QA** Dec '23 – Jan '23
*Supervisor: Self*

- **Google Colab** Document retrieval, extractive QA pipeline using RoBERTA in low-resource settings.
- Reduced retrieval + QA pipeline latency below 700 ms per query.

**Self-Supervised robust Adversarial Attack Detection** Jan '23 – Jan '23
*Supervisor: Prof. Richa Singh, Prof. Mayank Vatsa*

- **GitHub shrivastava95/dependable-ai** Implemented an Autoencoder to maximise the reconstruction error of adversarially attacked samples by implementing a self-supervised triplet loss objective by mimicing the behaviour of attacked samples.

## *Publications*

**Learn "No" to Say "Yes" Better: Improving Vision-Language Models via Negations (cited by 54)** *arXiv*
*Jaisidh Singh*, **Ishaan Shrivastava***, Mayank Vatsa, Richa Singh, Aparna Bharati* *Published - WACV2025*

## *Skills and Technologies*

**Machine Learning**: PyTorch, Jax, einops, Numpy, OpenCV, Ultralytics, HuggingFace, ONNX, Docker
**Programming Languages**: Python, Bash, C++/C, MATLAB, JavaScript
**Agentic Tools**: dsPy, LangGraph, LangChain, Codex, Claude Code, Hermes Agent

## *Coursework*

Machine Learning, Deep Learning, Artificial Intelligence, Dependable AI, Probability & Statistics, Stochastic Processes, Optimization for Machine Learning, Game Theory, Financial Engineering, Data Structures & Algorithms, Operating Systems, Computer Networks.

## *Positions of Responsibility*

**Core Member - Programming Club, IIT Jodhpur** August '22 – May '24

- Coordinator, Programming Club; Problemsetter, Tech Fest (Prometeo) at IIT Jodhpur

## *Other Interests*

Avid chess player, origami enthusiast, and active competitive programmer. I thrive when given clear, precise instructions and goals, and under the pressure of competition.

## *Achievements*

**Expert at Codeforces - Reached 'Expert' (rating 1749, 7 months, 33 contests) –** top 7 % on Codeforces.
All-India-Rank 1438 in JEE Mains 2020, All-India-Rank 3231 in JEE Advanced 2020. Was Awarded **International Rank 11** in SOF olympiad (2014)